

Schedule-Free On-Demand Water Transport in Bergen: Progress Report

[Author name]

Advisors: Julio Goez, Stein W. Wallace
NHH Norwegian School of Economics

August 31, 2026

Abstract

This report documents the progress made so far on a thesis project that designs the real-time dispatch logic for a small-boat, on-demand water transport service in Bergen, Norway. The service does not exist yet: rather than optimizing an existing system, the project builds the model of how such a service would operate, replacing fixed timetables with a dispatch policy that recomputes itself every few minutes. Three pieces are complete and verified at the time of writing: (i) a navigable routing module that computes real, coastline-respecting travel times between the four candidate stops, replacing an earlier straight-line approximation; (ii) a synthetic demand model anchored in official Norwegian population and employment data, combining a gravity-style formulation with a Poisson arrival process; and (iii) a discrete-time simulation environment, implemented to the Gymnasium interface standard, that models the operation of a small fleet of boats under a fixed reference dispatch policy. The environment has been verified on a small test instance: person-conservation holds exactly, and a coordination bug in the reference policy was found and fixed during verification, raising service compliance from 53% to 90% on the same run. This is a progress report, not a final one: a larger test instance and the reinforcement-learning agent that will be compared against the reference policy remain future work.

Contents

1	Introduction and Motivation	2
2	Related Work	2
3	Navigable Routing	3
4	Demand Model	4
5	Simulation Environment as a Markov Decision Process	7
5.1	State	7
5.2	Actions	7
5.3	Transition	8
5.4	Reward	8
6	Reference Dispatch Policy	8
7	Metrics and Results of an Example Run	9
8	Assumptions and Limitations	11
9	Progress Summary and Next Steps	12

1 Introduction and Motivation

Bergen is a coastal city whose public transport network is structured as a hub-and-spoke system centered on the city center, a layout imposed by the surrounding fjords and mountains. As the city grows and tourism increases, this network is under increasing pressure, and land-based expansion is constrained by the same geography that makes it congested in the first place. The fjords that divide the city are, at the same time, an underused transport corridor.

This project explores a small-boat, on-demand water transport service as a complementary mode. Two principles guide its design, established at the outset of the project and preserved throughout the work reported here. First, the service is not modeled as a profit-making venture: public transport of this kind is rarely self-financing, and its justification is environmental and related to congestion relief, not revenue. The objective is to deliver a good level of service efficiently, not to maximize income. Second, and more centrally to this report, the flexible layer of the service has no fixed timetable. A passenger requests a trip and a boat is dispatched to serve it, subject to a waiting-time guarantee, and the dispatch policy is recomputed at short, regular intervals rather than following a pre-published schedule. This "schedule-free" operating philosophy follows the line of work on schedule-free high-frequency transit control introduced by Sánchez-Martínez et al. [2017], who show that abandoning fixed timetables in favor of real-time headway and resource-aware control can outperform schedule-adherence strategies in high-frequency transit systems, provided a successful real-time control approach exists.

The reference point for this thesis is the water-taxi system studied by Gu and Wallace [2021] for the same four locations in Bergen used throughout this project (Kleppestø, Laksevåg, Bryggen, and Sandviken). That work formulates a *static* mixed-integer program that jointly decides terminal locations, fleet size, and routing for a representative demand pattern. The present project asks a different question: given a fleet and a set of stops, how should dispatch decisions be made *in real time*, minute by minute, as passengers actually arrive? This requires a simulation environment that can be driven by a dispatch policy and, eventually, by a learned agent – which is the piece of work documented in this report.

2 Related Work

Three papers anchor the modeling choices described in this report.

Gu and Wallace [2021] study a water-taxi system connecting the same four stops used here and formulate a static optimization model for terminal location, fleet sizing, and routing. It is the closest precedent to this project in terms of the physical instance (same stops, same city), and it supplies an independent, literature-based cross-check for the demand scale used in the simulation environment’s test instances (Section 4).

Braathen et al. [2024] study autonomous ferry scheduling under labor regulations for manned crews, comparing autonomous and manually operated fleets from a passenger-experience perspective. Two elements of that paper are adapted directly in this project: the multiplicative, gravity-style formula used to generate synthetic passenger demand (Section 4), and the reported statistics of passenger group sizes, which are used to calibrate the group-size distribution used here.

Sánchez-Martínez et al. [2017] develop the schedule-free operating paradigm for high-frequency transit referenced in the introduction. Their central argument – that abandoning fixed schedules in favor of real-time, resource-aware control can improve service quality, provided the underlying control problem can be solved effectively – is the conceptual motivation for treating the dispatch problem in this thesis as a control problem to be solved with a learned policy, rather than as a scheduling problem to be solved once at the planning stage.

3 Navigable Routing

Travel times between the four stops are computed from a navigable-routing module rather than from straight-line (Haversine) distances. An earlier, straight-line version of the travel-time matrix was found, on visual inspection, to draw lines that cross land near Bryggen, which sits inside the Vågen bay next to the Nordnes peninsula – straight-line distances are therefore not usable directly and are corrected here.

Water/land classification. A basemap tile mosaic covering the four stops is downloaded at zoom level 14, which corresponds to a fixed Web Mercator resolution of $156543.03392/2^{14} \approx 9.55$ m/pixel. Because Web Mercator distorts distances by a factor of $1/\cos(\text{latitude})$, and Bergen sits at approximately 60.4°N , the real ground resolution is $9.55 \times \cos(60.4^\circ) \approx 4.7$ m/pixel. Each pixel is classified as water or land by its Euclidean distance, in RGB space, to a reference water color sampled from the basemap style (threshold of 15 color units); the resulting mask is pixel-for-pixel aligned with the source map, since both are produced by the same Web Mercator projection formula and the same Earth radius.

Graph and shortest paths. Each water pixel becomes a node in an 8-connected graph (4 orthogonal and 4 diagonal neighbors), with edge weights equal to the real distance between pixel centers; land pixels are simply absent from the graph, so no edge can cross land. The four stops are snapped to the nearest water pixel, restricted to the graph’s main connected component, to avoid anchoring a stop to an isolated puddle unconnected to open water. A multi-source Dijkstra search (`scipy.sparse.csgraph.dijkstra`) then computes the shortest navigable path between every pair of stops in a single call, exact because all edge weights are non-negative real distances.

Design speed and resulting matrix. Rather than calibrating vessel speed against an existing ferry line, a fixed design speed of 30 km/h (≈ 16.2 knots) is used for the whole fleet, treated as a design decision for the small on-demand vessels rather than an empirical calibration. Table 1 reports the resulting travel-time matrix, and Table 2 compares it against the straight-line matrix it replaces: routing around land adds between 6% and 24% to travel distance depending on the pair, with Laksevåg – whose corrected location sits further inside the Puddefjorden – showing the largest correction because the detour around the Nordnes peninsula is large relative to its own distance to the other stops.

Table 1: Navigable travel-time matrix (minutes), 30 km/h design speed.

	Kleppestø	Laksevåg	Bryggen	Sandviken
Kleppestø	–	5.3	11.5	9.6
Laksevåg	5.3	–	8.6	8.6
Bryggen	11.5	8.6	–	6.8
Sandviken	9.6	8.6	6.8	–

Table 2: Straight-line vs. navigable travel times, all six stop pairs.

Pair	Straight-line (min)	Navigable (min)	Difference
Laksevåg ↔ Bryggen	7.0	8.6	+23.8%
Kleppestø ↔ Sandviken	8.7	9.6	+10.5%
Bryggen ↔ Sandviken	6.3	6.8	+8.9%
Kleppestø ↔ Laksevåg	4.9	5.3	+7.8%
Kleppestø ↔ Bryggen	10.7	11.5	+7.3%
Laksevåg ↔ Sandviken	8.1	8.6	+6.4%

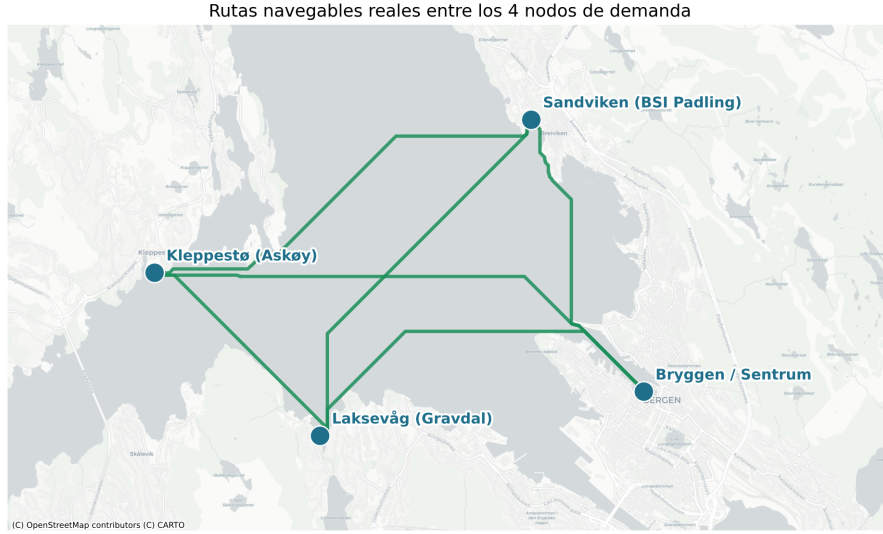


Figure 1: The six navigable routes between the four demand stops, computed by Dijkstra search over the water-classified pixel graph.

The known limitation of this method is that an 8-connected grid only offers eight travel directions per step, so a shortest path can zig-zag slightly instead of following a perfect diagonal when the true direction falls between two of the eight allowed ones; the resulting overcost is a few percent in the worst case and does not affect the central conclusion that the method reliably avoids land.

4 Demand Model

No real demand data exists for this service, since it does not operate yet. Demand is therefore synthesized, but anchored in real open data wherever possible, following the principle that invented parameters should be declared explicitly rather than hidden inside the code.

Catchment areas and masses. Population and employment are read from official Statistics Norway (SSB) 250-metre grids. For each of the four stops, a catchment area was hand-drawn by the project team in QGIS; every *grunnkrets* (Norway’s smallest statistical reporting unit) that the drawn area touches, even partially, is included in full for that stop, and duplicate assignments at shared boundaries are resolved by keeping each *grunnkrets* with the stop it overlaps the most. Population and employment are then summed over the SSB grid cells inside the resulting area. Table 3 reports the resulting masses.

Table 3: Population and employment mass by node (SSB grids, 2026).

Node	Area (km ²)	Population	Employment
Kleppestø (Askøy)	80.6	25,153	7,550
Laksevåg (Gravdal)	14.9	15,986	7,760
Bryggen / Sentrum	20.2	27,695	52,213
Sandviken (BSI Padling)	14.6	11,109	4,106

Origin-destination pattern. Following the multiplicative, gravity-style formulation of Braathen et al. [2024] (their Section 4.1), the popularity of an origin-destination pair at a given hour is computed as

$$\text{pop}_{pg} := \ln \left(1 + \prod_{i=1}^4 p_{i,pg} \right), \quad (1)$$

where the four factors p_i are, in the original paper, station-level popularity scores set by judgment for origin, destination, weekday, and hour. This project’s contribution is to replace the origin and destination popularity scores with the real SSB masses of Table 3 (population when the node plays the role of a trip origin, employment when it plays the role of a trip destination, inverted for the afternoon), and to add a distance-decay factor

$$f_{\text{distance}}(t) = \exp(-t/\tau), \quad \tau = 15 \text{ min}, \quad (2)$$

where t is the navigable travel time from Section 3 – an addition not present in the original formulation. Table 4 lists the four time bands used, with their relative volume factor and the role each mass plays; the resulting morning peak concentrates 85.6% of its intensity toward Bryggen, close to the 90/10 split assumed *a priori* before this model existed, but now derived from the data rather than assumed.

Table 4: Time bands and relative demand volume.

Band	Hours	Volume factor	Origin / destination role
Morning	06–09	1.00	population $\rightarrow$ employment
Midday	09–15	0.40	mixed (population/employment average)
Afternoon	15–18	1.00	employment $\rightarrow$ population
Evening	18–24	0.15	mixed

Arrivals and group sizes. Passenger arrivals are drawn from a Poisson process per origin-destination pair and hour, at a rate proportional to the intensity computed above. Braathen et al. [2024] report a mean group size of 8.6 passengers with standard deviation 2.6, but do not specify a sampling distribution (their own group size is a deterministic function of a popularity score). Since group size is a count of people, a discrete, non-negative distribution is more appropriate than a continuous one; this project uses a Binomial($n = 40, p = 0.215$) distribution, calibrated by matching moments to Braathen et al.’s reported mean and standard deviation, which reproduces the target mean exactly (8.60) and the target standard deviation closely (2.598 against 2.6).

Demand scale. Neither reference paper reports a modal-share percentage directly. One is inferred by cross-referencing their reported demand scales against the population of the four catchment areas: Gu and Wallace [2021], who study the same four stops, assume 300 travel requests per day, which – combined with Braathen et al.’s group-size statistics – corresponds to about 3.2% of the catchment population; Braathen et al. [2024]’s larger test instance, for a

hypothetical six-stop citywide network, corresponds to about 4.3% of Bergen’s total population. Both cross-checks converge on a 3–4% range. The demand instance used in this project sets this parameter at 10%, deliberately above that range, representing a higher-adoption scenario rather than a direct reading of either reference – a decision the team makes explicitly, not a literature-derived number, and declared as such.

Persons vs. groups. Braathen et al. [2024]’s group is the atomic unit of the demand-generation process above. For the dispatch simulation environment (Section 5), each group is optionally exploded into individual persons, each inheriting the same origin, destination, arrival time, and patience as the group they came from – this changes the atomic unit seen by the simulator’s capacity and boarding logic without altering when or where demand appears.

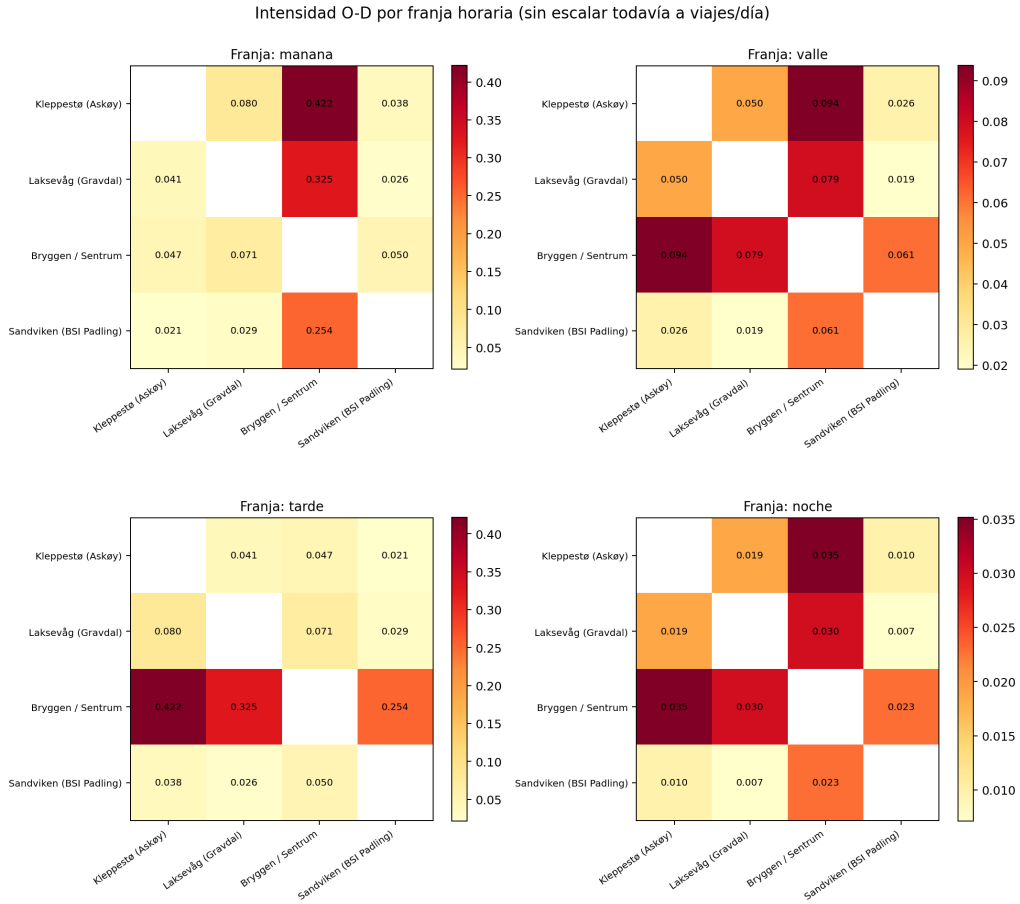


Figure 2: Origin-destination intensity heatmaps by time band. Morning traffic concentrates toward Bryggen; afternoon reverses the pattern.

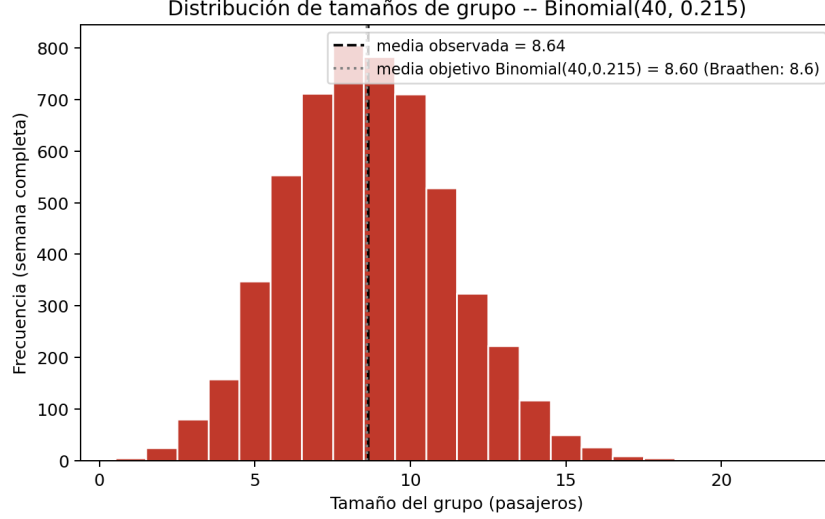


Figure 3: Observed group-size distribution over a simulated week, against the target Binomial(40, 0.215).

5 Simulation Environment as a Markov Decision Process

The dispatch problem is modeled as a Markov decision process and implemented as a discrete-time simulation environment following the Gymnasium interface standard (`reset()` / `step()`), which is not a library of pre-built simulators but an interface contract: the environment implements the entire boat-and-passenger logic itself, and only its shape is standardized, so that a reinforcement-learning library can later drive it without modification. The environment advances in **fixed 2-minute steps**, rather than jumping between discrete events, for simplicity and because it maps directly onto the `step()` call.

5.1 State

The state has three parts. For each boat: origin node, destination node (equal to the origin if idle), minutes remaining until arrival, current occupancy, and whether it is free to receive a new order. For demand: one queue per origin-destination pair (twelve pairs for four nodes) – not one queue per node – holding the number of people waiting and the waiting time of the oldest person in that specific pair. For time: minute of day and day of week. This rich, dictionary-shaped state is used for logging and inspection; a separate fixed-size numeric vector is derived from it for the eventual learning agent, using one-hot node encoding and cyclical (sine/cosine) time encoding so that midnight and day-of-week wraparound do not appear as artificial discontinuities to a neural network.

5.2 Actions

Every 2 minutes, each free boat receives one of five orders: travel to one of the four nodes, or wait. A boat already en route ignores any new order and continues to its destination; it is not redirected mid-trip. Boarding follows a fixed rule, not chosen by the policy: when a free boat at node A is ordered to travel to node B , it boards strictly from the queue for the pair (A, B) , in arrival order, up to capacity, and travels directly to B , where everyone aboard disembarks. Trips are therefore always direct, point-to-point journeys between exactly two stops; a boat never carries passengers with mixed destinations, since it can only ever board from the one queue that matches the trip it was just ordered to make.

One behavior not specified explicitly in the original design was resolved during implementation: if the most urgent waiting passengers are at a node other than the boat’s current position, the boat repositions there empty first (this step’s action carries no boarding), and only on its *next* free moment – once actually at that node, and re-evaluating with fresh information – does it consider boarding and departing. This is a deliberately myopic, greedy rule, consistent with the reference policy being a simple heuristic rather than a two-step planner.

5.3 Transition

Each step executes, in order: (1) apply the chosen action to every free boat, boarding from the matching pair queue as described above; (2) advance all boats in transit by the fixed time step, disembarking everyone aboard any boat that reaches its destination; (3) add to the queues any new arrivals whose arrival time falls within the interval just elapsed – these do not board the boat that just departed in step (1), since they had not yet arrived when boarding happened; (4) remove, as lost, any waiting person whose patience has been exceeded; (5) compute the step’s reward and assemble the new state.

5.4 Reward

The reward is negative (a sum of penalties) and combines three terms, computed every 2 minutes for every person still active in the system – waiting *or* aboard a boat not yet arrived, since the specification defines this term over every passenger who has not yet reached their destination:

$$\text{surplus} = \max(0, \text{time in system} - \tau_{\text{tol}}), \quad \text{surplus}_{\text{max}} = \max(\varepsilon, \text{patience} - \tau_{\text{tol}}), \quad (3)$$

$$r = - \left[\sum_{\text{active persons}} \left(\frac{\text{surplus}}{\text{surplus}_{\text{max}}} \right)^2 + P_{\text{lost}} \cdot (\text{persons lost this step}) + w_{\text{move}} \cdot (\text{boats under way this step}) \right], \quad (4)$$

with tolerance $\tau_{\text{tol}} = 12$ minutes (the longest direct navigable route, from Table 1), loss penalty $P_{\text{lost}} = 1.3$, and movement penalty $w_{\text{move}} = 0.1$; all three parameters live in the environment’s configuration file, not hard-coded.

A gap in the specification was found and closed during verification: patience on strong connections is 15 minutes with random jitter of ± 5 minutes (already part of the demand generator), so it can fall as low as 10 minutes – below the 12-minute tolerance, which would make $\text{surplus}_{\text{max}}$ zero or negative for those cases. This is corrected with the floor $\varepsilon = 1.0$ minute shown above: for the rare cases this affects, the penalty simply saturates near its maximum almost immediately, which is a reasonable outcome for someone about to be lost. The demand generator and its jitter were left untouched; the correction lives entirely in the reward function.

6 Reference Dispatch Policy

Before any learning is introduced, the environment is driven by a fixed, “dumb” reference policy – a nearest-available rule – both to verify that the environment behaves correctly and to later serve as the baseline the learned agent must beat. When a boat becomes free, it is sent to serve the most urgent demand it can still reach in time, prioritizing demand departing from its own current node (servable immediately) over repositioning toward demand elsewhere (Section 5).

Applying this rule independently to each free boat, using the same pre-step snapshot of the world for all of them, produced a coordination bug: two boats free at the same node and moment could both decide to serve the same destination, since neither’s decision accounted for the other’s simultaneous choice. The first boat processed would take all the waiting passengers; the second would travel to the same destination empty, paying the movement penalty while serving no one. This was found by inspecting a single simulated minute in detail (Section 7)

and confirmed in the step-by-step log. It was corrected by coordinating the free boats of a given step sequentially: each boat’s decision is computed against a locally updated copy of the queues that accounts for what the boats already assigned this round would take, so a queue that the first boat would empty is correctly seen as empty by the second. The measured effect of this fix, on the example run reported in Section 7, was substantial: service compliance rose from 53% to 90% on the identical instance.

7 Metrics and Results of an Example Run

The environment and reference policy were verified on a small test instance (“Escalón 1”): the morning band (06:00–09:00), two boats of capacity 20, and demand scaled down from the demand model of Section 4 (0.8% of the catchment population instead of the model’s own 10%, chosen only so the instance is small enough to inspect by hand) – 23 passenger groups, exploded into 202 individual persons.

Conservation check. Every person generated must end the run in an accountable state: served, lost, or still waiting/aboard if the run was truncated mid-trip. This is checked explicitly and reported, not assumed: of 202 persons generated, 181 were served, 21 were lost, and none remained waiting or aboard at the end of the run – the counts add up exactly.

Table 5: Global metrics, example run (Escalón 1).

Metric	Value
Overall compliance (served / generated)	89.6%
Persons served / lost	181 / 21
Mean waiting time	10.7 min
Mean / maximum time in system	19.5 / 34.0 min

Results vary substantially by origin-destination pair: with only two boats, some directions are served perfectly (100% compliance on, e.g., Kleppestø→Sandviken and Kleppestø→Bryggen) while others are underserved (Bryggen→Laksevåg at 43.8%, Sandviken→Bryggen at 58.6%) – a realistic and expected consequence of a small fleet, not a modeling artifact.

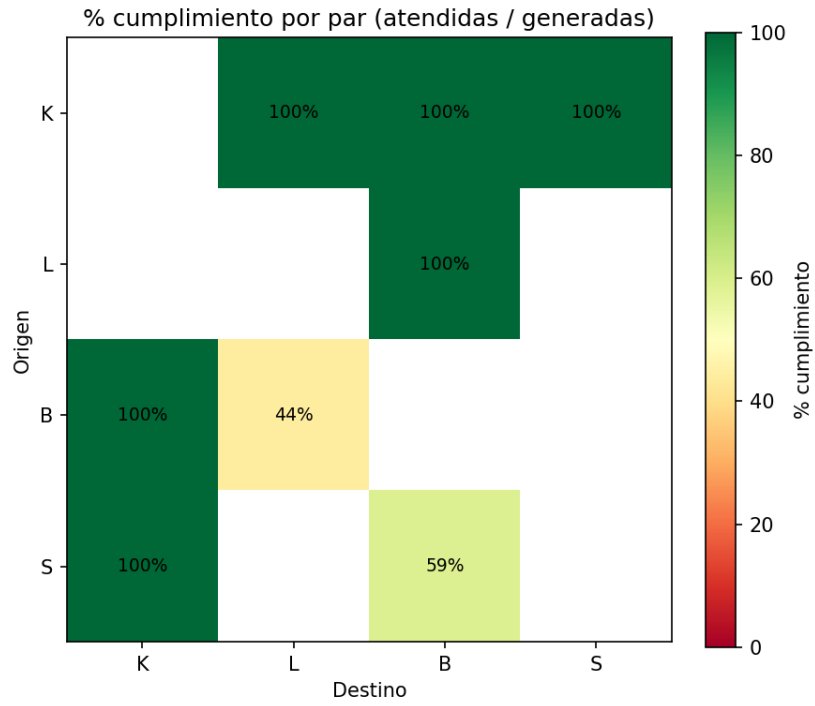


Figure 4: Compliance rate (served / generated) by origin-destination pair.

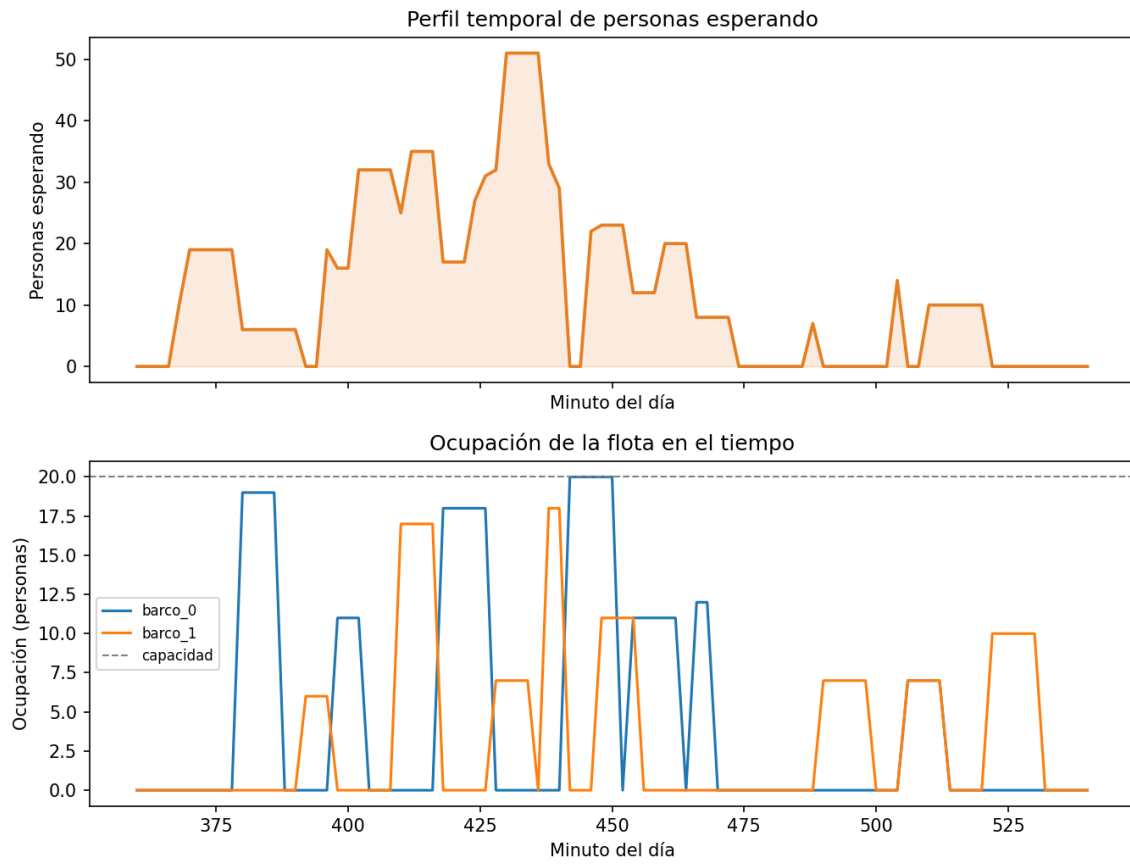


Figure 5: Waiting queue length (top) and fleet occupancy (bottom) over the simulated morning.

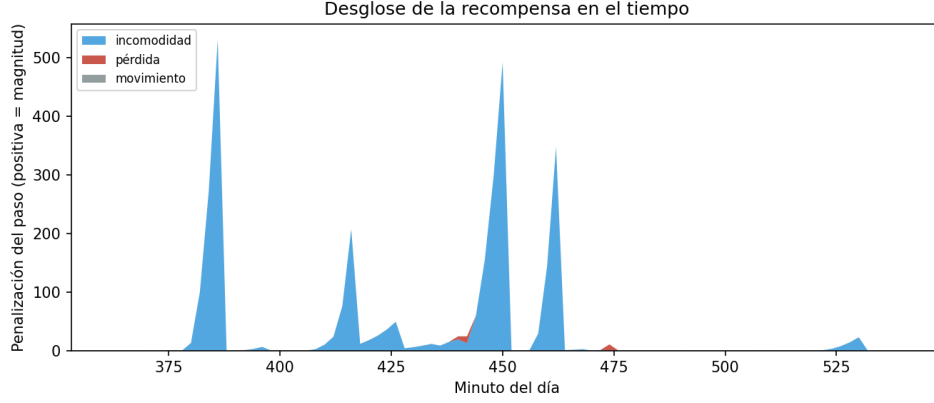


Figure 6: Reward breakdown by term over the simulated morning.

Inspecting a single minute. Beyond aggregate metrics, the environment supports pausing at an exact simulated minute and displaying the full state – boat positions and occupancy, every non-empty queue, the decision each boat made that step, and the reward breakdown – both as text and as a map. Figure 7 shows minute 400 (06:40) of the example run, the exact minute at which the fleet-coordination bug described in Section 6 was found.



Figure 7: State inspector, minute 400 of the example run.

The environment also produces an animation of the full run over the real navigable routes of Section 3, with boats interpolated along the actual coastline-respecting paths rather than straight lines, and queue counters that grow and shrink at each stop; it is not reproduced here as a static figure since it is inherently a video artifact.

A larger test instance ("Escalón 2": a full day, three boats) is designed but not yet executed at the time of this report; it is listed as the immediate next step in Section 9.

8 Assumptions and Limitations

This section lists the assumptions carried forward from the work described above, ordered roughly by how much leverage each has on the final results.

- **Hand-drawn catchment areas.** The single largest source of downstream variation: another reasonable drawing of the four catchment areas would change population and

employment masses, and therefore the demand scale, which is itself a percentage of that population.

- **Demand scale (10% of catchment population).** Anchored in a 3–4% range cross-referenced from two independent papers, but the 10% actually used is a deliberate team decision above that range, not a literature-derived figure.
- **University population factor.** Bergen’s registered employment figures do not capture students, and no open source of fine-grained enrollment data was found; a manual weighting factor compensates for this at the two nodes furthest from campuses, declared explicitly as a design choice rather than measured data.
- **Group-size distribution (Binomial(40, 0.215)).** Calibrated to match the mean and standard deviation reported by Braathen et al. [2024], not their deterministic popularity-based formula, which depends on a scale parameter specific to their own station data.
- **Test-instance demand scale (0.8% / 1.2% of population for Escalón 1 / 2).** Chosen only so the verification instances are small enough to inspect by hand; unrelated to the 10% figure used for the ”official” demand model.
- **Reward tolerance and loss-penalty floor.** The 12-minute comfort tolerance and the $\varepsilon = 1$ minute floor on $\text{surplus}_{\max}$ are, respectively, a specification parameter and a correction found necessary during verification (Section 5.4); neither is calibrated against passenger behavior data, since the service does not exist.
- **The reference policy is myopic and greedy,** not an optimizer: it never plans more than one leg ahead, and the fleet-coordination fix (Section 6) resolves same-step collisions between boats but still assigns them sequentially in a fixed order, not by evaluating the jointly optimal assignment.
- **The simulation environment itself has no internal randomness.** All stochasticity in the pipeline lives in the demand generator, which is separately seeded; once a demand table is fixed, the environment is fully deterministic, which was treated as a desirable property for verification rather than a limitation.

9 Progress Summary and Next Steps

At the time of this report, three pipeline stages are complete and verified: navigable routing (Section 3), the synthetic demand model (Section 4), and the discrete-time simulation environment together with its reference dispatch policy (Sections 5–7). Person-conservation holds exactly on the verification run, and the two implementation issues found along the way – a reward-formula edge case and a fleet-coordination bug – were identified through explicit verification (a conservation check, a step-by-step log, and a state inspector), fixed, and are documented here rather than hidden.

The immediate next steps are: (i) run and report the larger ”Escalón 2” instance (a full day, three boats) to obtain metrics at a more representative scale; (ii) connect a reinforcement-learning agent – planned as Deep Q-Networks via Stable-Baselines3, per the environment’s Gymnasium interface – and train it on the same small instances used to verify the reference policy; (iii) compare the learned policy against the nearest-available baseline reported here, and, at larger scale, against a static optimization benchmark in the spirit of Gu and Wallace [2021]. The contribution of the thesis is concentrated in this last comparison: a real-time, learned dispatch policy for a schedule-free system, a control problem that the static models in the literature reviewed in Section 2 do not address.

References

- Caroline Braathen, Julio C. Goetz, and Mario Guajardo. Autonomous ferries in light of labor regulations – a passenger perspective. *Maritime Transport Research*, 7:100115, 2024. doi: 10.1016/j.martra.2024.100115.
- Yong Gu and Stein W. Wallace. Operational benefits of autonomous vessels in logistics – a case of autonomous water-taxis in Bergen. *Transportation Research Part E: Logistics and Transportation Review*, 154:102456, 2021. doi: 10.1016/j.tre.2021.102456.
- Gabriel E. Sánchez-Martínez, Nigel H. M. Wilson, and Haris N. Koutsopoulos. Schedule-free high-frequency transit operations. *Public Transport*, 9:285–305, 2017. doi: 10.1007/s12469-016-0129-8.